PRISM: Mathematical Foundations

Sharpe–Optimal Allocation under Forecast Uncertainty
with Kantorovich Order Flow

Benjamin Coriat
McGill University — ECSE 551

April 2026

Abstract

We derive from first principles the optimal portfolio allocation rule when the investor possesses *noisy* return and covariance forecasts together with a full characterisation of the forecast errors. The key idea is that the classical mean–variance programme is mis-specified when the inputs are themselves uncertain: we show how to build a *confidence-weighted* expected return vector and a *Bayesian-blended* covariance matrix, then prove that the resulting Sharpe-maximising programme is a second-order cone programme. Finally, we derive the Kantorovich optimal transport plan that converts the current portfolio into the target at minimum transaction cost, and show that its entropically regularised form admits a closed-form Sinkhorn iteration.

Contents

1 Setup and Data **3**

2 The Forecasting Step (Black Box) **3**

3 Error Characterisation **3**

4 From Error Statistics to Confidence **4**
4.1 Designing the confidence score 4
4.2 Confidence-weighted returns 5

5 Bayesian Posterior Estimation **5**
5.1 Prior specification 5
5.2 Posterior update 5

6 Robust Covariance Construction **6**
6.1 EWMA volatility 6
6.2 Regime adjustment 6
6.3 Final blended covariance 7

7 Sharpe-Optimal Allocation **7**
7.1 The classical Markowitz problem and its flaw 7
7.2 The constrained programme 7
7.3 Deriving the KKT conditions 8

8 Volatility Targeting **8**

9 Optimal Transport for Order Execution **8**
9.1 The Kantorovich formulation 8
9.2 Entropic regularisation 9
9.3 Sinkhorn algorithm 9
9.4 Transaction cost and net return 10

10 Performance Metrics **10**

1 Setup and Data

Fix a universe of n assets and a weekly time index t (Friday close).

Definition 1.1 (Returns and volatility)**.** *For asset i at week t:*

$$r_{i,t} := \frac{P_{i,t}}{P_{i,t-1}} - 1 \qquad \textit{(simple weekly return)}, \tag{1}$$

$$\hat{\sigma}^{\mathrm{GK}}_{i,t} := \sqrt{\max\Big(\tfrac{1}{2}(\ln H - \ln L)^2 - (2\ln 2 - 1)(\ln C - \ln O)^2,\ 0\Big)} \qquad \textit{(Garman–Klass vol)}. \tag{2}$$

Remark 1.2 (Why Garman–Klass?)**.** *Close-to-close volatility uses only two prices per period. The GK estimator uses all four OHLC prices; under driftless GBM one can show* $\mathrm{Var}(\hat{\sigma}^2_{\mathrm{GK}}) \approx 0.273\,\sigma^4$*, roughly* $7.3\times$ *more efficient. Better volatility estimates feed directly into a better covariance matrix, so this matters for allocation.*

Definition 1.3 (Rolling correlation)**.** *With window $W = 26$,*

$$\hat{\rho}_{ij,t} := \frac{\sum_{s=t-W+1}^{t}(r_{i,s} - \bar{r}_i)(r_{j,s} - \bar{r}_j)}{\sqrt{\sum_s (r_{i,s} - \bar{r}_i)^2 \sum_s (r_{j,s} - \bar{r}_j)^2}}.$$

Collect these into the correlation matrix $\hat{C}_t \in \mathbb{R}^{n\times n}$.

2 The Forecasting Step (Black Box)

A stacked ensemble regressor (Ridge, GBM, Random Forest, MLP, etc.) produces, at each date t:

1. A predicted return vector $\hat{\boldsymbol{\mu}}_{t+1} \in \mathbb{R}^n$.
2. A predicted correlation matrix $\hat{C}_{t+1} \in \mathbb{R}^{n\times n}$.

The details of the ensemble are not the focus here. What matters is that these forecasts are *imperfect*, and we have a way to measure how imperfect they are — the **error characterisation** described next.

3 Error Characterisation

For each asset i, define the signed forecast error at past date s:

$$\varepsilon_{i,s} := \hat{r}^{\mathrm{pred}}_{i,s} - r_{i,s+1}.$$

We maintain a running history $\mathcal{H}_{i,t} = \{\varepsilon_{i,s}\}_{s<t}$ and compute five statistics that together describe *how trustworthy* the forecast for asset i is.

Definition 3.1 (Return error matrix $E_\mu \in \mathbb{R}^{n\times 5}$)**.**

$$[E_\mu]_{i,1} := \sum_s w_s\,\varepsilon_{i,s} \qquad \textit{EWMA bias } (\lambda = 0.94), \tag{3}$$

$$[E_\mu]_{i,2} := \mathbf{1}[\mathrm{sign}(\hat{r}_{i,t-1}) = \mathrm{sign}(r_{i,t})] \qquad \textit{last directional hit}, \tag{4}$$

$$[E_\mu]_{i,3} := \mathrm{std}(\mathcal{H}_{i,t}) \qquad \textit{error volatility}, \tag{5}$$

$$[E_\mu]_{i,4} := \mathrm{skew}(\mathcal{H}_{i,t}) \qquad \textit{error skewness}, \tag{6}$$

$$[E_\mu]_{i,5} := \mathrm{exkurt}(\mathcal{H}_{i,t}) \qquad \textit{error excess kurtosis}. \tag{7}$$

Remark 3.2 (Intuition)**.** *Column 1 tells us whether the model is* systematically *over- or under-predicting (and recent errors are up-weighted). Column 2 says whether it at least gets the* sign *right. Columns 3–5 describe the* shape *of the error distribution: a wide, left-skewed, heavy-tailed error distribution means we should trust the forecast less.*

Similarly, for each pair (i, j) we track $\varepsilon^{C}_{ij,s} := \hat{\rho}^{\text{pred}}_{ij,s} - \hat{\rho}_{ij,s+1}$ and store bias, std, skewness, kurtosis in a tensor $E_C \in \mathbb{R}^{n\times n\times 4}$.

Finally, the last $k = 26$ weeks of GK volatilities are stored as $V \in \mathbb{R}^{n\times k}$.

4 From Error Statistics to Confidence

The central question: given E_μ, how should we *down-weight* the return forecasts we don't trust?

4.1 Designing the confidence score

We want a score $\kappa_i \in (0, 1]$ for each asset, where $\kappa_i = 1$ means "fully trust the forecast" and $\kappa_i \approx 0$ means "ignore it." We build κ_i multiplicatively from four penalty terms, each in $(0, 1]$.

Design principle

Each penalty is an exponential decay in a normalised badness measure. If an error statistic is at the *median* level, the penalty is moderate; if it is many MADs above median, the penalty drives κ_i toward zero.

Definition 4.1 (Component scores)**.** *Let* $b_i := [E_\mu]_{i,1}$, $d_i := [E_\mu]_{i,2}$, $v_i := [E_\mu]_{i,3}$, $\xi_i := [E_\mu]_{i,4}$, $\kappa^{\text{ex}}_i := \max([E_\mu]_{i,5}, 0)$.

1. Bias penalty. *Large systematic bias ⇒ low trust:*

$$s^{\text{bias}}_i := \exp\Big(- \alpha_1 \frac{|b_i|}{\text{MAD}(\boldsymbol{b})}\Big), \qquad \text{MAD}(\boldsymbol{b}) := \max\big(\text{median}(|b_j - \text{median}(\boldsymbol{b})|),\, 10^{-8}\big). \tag{8}$$

Why MAD? *The median absolute deviation is robust to outliers, so one rogue asset doesn't inflate the scale and mask problems in others.*

2. Directional accuracy. *If the model can't even get the sign right, down-weight:*

$$s^{\text{dir}}_i := \beta_1\, d_i + (1 - \beta_1), \qquad \beta_1 = 0.3. \tag{9}$$

This is simply a linear interpolation: a correct sign gives $s^{\text{dir}} = 1$*, a miss gives* $1 - \beta_1 = 0.7$*. The penalty is intentionally mild because a single wrong sign shouldn't nuke the confidence.*

3. Dispersion penalty. *High error volatility, especially combined with fat tails, means the forecast is unreliable:*

$$\delta_i := v_i\,(1 + \gamma_1 \kappa^{\text{ex}}_i), \qquad s^{\text{disp}}_i := \exp\Big(- \alpha_2 \frac{\delta_i}{\text{median}(\boldsymbol{\delta})}\Big). \tag{10}$$

The term $\gamma_1 \kappa^{\text{ex}}_i$ *inflates the effective dispersion when the error distribution has excess kurtosis — because fat tails mean the standard deviation* underestimates *the true forecast risk.*

4. Skewness penalty. *Negatively skewed errors (model over-predicts) are penalised:*

$$s^{\text{skew}}_i := \exp\big(- \alpha_3 \max(-\xi_i, 0)\big). \tag{11}$$

Only negative *skew is penalised: over-prediction is worse than under-prediction for a long-biased portfolio.*

Definition 4.2 (Composite confidence)**.**

$$\kappa_i := \mathrm{clip}\Big(\prod_{k=1}^{4} \big(s_i^{(k)}\big)^{\omega_k},\ 10^{-4},\ 1\Big), \qquad \omega_k = \tfrac{1}{4}. \tag{12}$$

Remark 4.3 (Why geometric mean?)**.** *A multiplicative structure means that* any single *bad signal can substantially reduce confidence. If we used an arithmetic mean, a terrible bias score could be offset by a good directional score, which is undesirable: we want a conjunction ("trust only if* all *diagnostics are acceptable").*

4.2 Confidence-weighted returns

Armed with κ_i, we shrink each forecast toward the cross-sectional mean:

$$\tilde{\mu}_i := \kappa_i\, \hat{\mu}_i + (1 - \kappa_i)\, \mu_0, \qquad \mu_0 := \frac{1}{n} \sum_{j=1}^{n} \hat{\mu}_j. \tag{13}$$

Remark 4.4 (Interpretation)**.** *When $\kappa_i = 1$, we fully believe the forecast. When $\kappa_i \to 0$, the asset's expected return is pulled to the market average, reflecting maximum uncertainty ("we know nothing special about this asset"). This is a form of* James–Stein shrinkage*: it provably reduces expected squared error relative to the raw forecasts when $n \geq 3$.*

5 Bayesian Posterior Estimation

The confidence-weighted returns $\tilde{\boldsymbol{\mu}}$ are a point estimate. We now embed them in a Bayesian framework to produce a *regularised* mean and covariance that account for estimation uncertainty.

5.1 Prior specification

Assumption 5.1 (Normal–Inverse-Wishart conjugate prior)**.**

$$\Sigma \sim \mathcal{IW}(\nu_0, \Psi_0), \qquad \nu_0 = n + 2, \tag{14}$$

$$\boldsymbol{\mu} \mid \Sigma \sim \mathcal{N}(\boldsymbol{m}_0,\, \tau_0^{-1}\Sigma), \qquad \tau_0 = 1, \quad \boldsymbol{m}_0 = \mu_0 \mathbf{1}. \tag{15}$$

The scale matrix is $\Psi_0 = \max(\nu_0 - n - 1, 1)\, \hat{\Sigma}_{\mathrm{hist}}$, where $\hat{\Sigma}_{\mathrm{hist}}$ is the historical sample covariance on the training period.

Remark 5.2 (Why these choices?)**.** *Setting $\nu_0 = n + 2$ gives the Inverse-Wishart the minimum number of degrees of freedom for which the mean exists — i.e. a* diffuse *prior that doesn't impose strong beliefs. $\tau_0 = 1$ means the prior on $\boldsymbol{\mu}$ has the same weight as a single observation, so the data quickly dominates. $\boldsymbol{m}_0 = \mu_0 \mathbf{1}$ centres the prior on equal expected returns (maximum ignorance).*

5.2 Posterior update

We treat the confidence-weighted vector $\tilde{\boldsymbol{\mu}}$ as a single "observation" from $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, but re-weighted by the confidence matrix $K = \mathrm{diag}(\kappa_1, \ldots, \kappa_n)$.

Proposition 5.3 (Posterior point estimates)**.** *The posterior mean and covariance mode are:*

$$\boldsymbol{\mu}_{t+1}^{B} = \frac{\tau_0\, \boldsymbol{m}_0 + K\tilde{\boldsymbol{\mu}}}{\tau_0 + 1}, \tag{16}$$

$$\Sigma_{t+1}^{B} = \frac{\Psi_n}{\max(\nu_0 + 1 - n - 1,\, 1)}, \tag{17}$$

where $\Psi_n := \Psi_0 + \frac{\tau_0}{\tau_0+1}(\tilde{\boldsymbol{\mu}} - \boldsymbol{m}_0)(\tilde{\boldsymbol{\mu}} - \boldsymbol{m}_0)^\top$.

Derivation. Under the NIW conjugate model with prior $(\boldsymbol{m}_0, \tau_0, \Psi_0, \nu_0)$, the standard update after observing data $\bar{\boldsymbol{x}}$ with effective sample size n_{eff} gives posterior parameters $\boldsymbol{m}_n = \frac{\tau_0 \boldsymbol{m}_0 + n_{\text{eff}}\bar{\boldsymbol{x}}}{\tau_0 + n_{\text{eff}}}$. Here we have a single "observation" ($n_{\text{eff}} = 1$) but it is modulated by the confidence matrix K: assets with $\kappa_i \approx 1$ contribute their forecast at full strength, while low-confidence assets are effectively shrunk toward the prior mean μ_0.

The scale matrix update $\Psi_n = \Psi_0 + \frac{\tau_0}{\tau_0+1}(\tilde{\boldsymbol{\mu}} - \boldsymbol{m}_0)(\tilde{\boldsymbol{\mu}} - \boldsymbol{m}_0)^\top$ is the standard NIW scatter update: it inflates the posterior covariance in directions where the observation deviates from the prior mean, reflecting the additional uncertainty from disagreement between prior and data.

The result is projected onto the positive-definite cone via Higham alternating projections (nearest correlation/covariance matrix). □

6 Robust Covariance Construction

The Bayesian posterior Σ_{t+1}^{B} handles mean uncertainty. We also need a *forward-looking* covariance from the predicted correlation matrix and recent volatilities.

6.1 EWMA volatility

From the volatility history $V \in \mathbb{R}^{n\times k}$ ($k = 26$), with decay $\lambda_v = 0.94$:

$$\hat{\sigma}_i := \sqrt{\sum_{\ell=1}^{k} w_\ell\, V_{i,\ell}^2}, \qquad w_\ell = \frac{(1-\lambda_v)\lambda_v^{k-\ell}}{1-\lambda_v^k}. \tag{18}$$

Remark 6.1. *The EWMA weights recent observations exponentially more.* $\lambda_v = 0.94$ *gives a half-life of about* $\ln 2/\ln(1/0.94) \approx 11$ *weeks, so the estimate adapts to regime changes within a quarter.*

6.2 Regime adjustment

If current vol is elevated relative to its own k-week average, we further inflate it:

$$\hat{\sigma}_i^{\text{adj}} := \hat{\sigma}_i \cdot \big(1 + \eta \max(r_i^{\text{vol}} - 1, 0)\big), \qquad r_i^{\text{vol}} := \hat{\sigma}_i/\bar{V}_i, \quad \eta = 0.5. \tag{19}$$

This is a simple asymmetric adjustment: vol is only inflated (never deflated), which biases toward conservatism during stress.

6.3 Final blended covariance

The forward-looking covariance uses the predicted correlation $\hat{C}_{t+1}$ and adjusted vols:

$$\Sigma^{\text{fwd}} := D_\sigma \, \hat{C}_{t+1} \, D_\sigma, \qquad D_\sigma = \text{diag}(\hat{\sigma}_1^{\text{adj}}, \ldots, \hat{\sigma}_n^{\text{adj}}). \tag{20}$$

We blend with the Bayesian posterior:

$$\boxed{\Sigma_{\text{final}} := \zeta \, \Sigma_{t+1}^B + (1-\zeta) \, \Sigma^{\text{fwd}}, \qquad \zeta = 0.5.} \tag{21}$$

This is a convex combination of two PSD matrices, hence PSD. We project to strictly PD (eigenvalue floor) for numerical stability.

7 Sharpe-Optimal Allocation

7.1 The classical Markowitz problem and its flaw

The textbook approach maximises the Sharpe ratio:

$$\text{SR}(\boldsymbol{w}) := \frac{\boldsymbol{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\boldsymbol{w}^\top \Sigma \boldsymbol{w}}}. \tag{22}$$

The well-known difficulty is that the solution $\boldsymbol{w}^* \propto \Sigma^{-1}\boldsymbol{\mu}$ is *extremely sensitive* to errors in $\boldsymbol{\mu}$ and Σ. Small perturbations in the inputs produce wildly different portfolios (Michaud, 1989).

Our remedy

By replacing $\boldsymbol{\mu}$ with the confidence-shrunk, Bayesian-updated $\boldsymbol{\mu}_{t+1}^B$ and Σ with the blended Σ_{final}, we have already regularised the inputs. The optimiser now sees "de-noised" inputs whose estimation error has been attenuated. This is the payoff of Sections 4–6.

7.2 The constrained programme

We solve, for each risk-aversion $\gamma > 0$:

$$\boldsymbol{w}^*(\gamma) := \arg\max_{\boldsymbol{w} \in \mathbb{R}^n} \underbrace{(\boldsymbol{\mu}_{t+1}^B - r_f \mathbf{1})^\top \boldsymbol{w}}_{\text{expected excess return}} - \gamma \underbrace{\boldsymbol{w}^\top \Sigma_{\text{final}} \boldsymbol{w}}_{\text{portfolio variance}} \tag{23}$$

subject to

$$\mathbf{1}^\top \boldsymbol{w} = 1 \qquad \text{(fully invested)}, \tag{24}$$

$$\|\boldsymbol{w}\|_1 \le L_{\max} = 1.6 \qquad \text{(gross leverage cap)}, \tag{25}$$

$$|w_i| \le w_{\max} = 0.20, \ \forall i \qquad \text{(concentration limit)}. \tag{26}$$

Remark 7.1 (Why not maximise SR directly?)**.** *Maximising* $\text{SR}(\boldsymbol{w})$ *is a fractional programme. The standard trick (Cornuejols & Tütüncü) is to fix the denominator and sweep* γ*. For each* γ*,* (23) *is a convex QP (since* $\Sigma_{\text{final}} \succ 0$*), solvable in milliseconds by SCS. We then pick the* γ^* *that achieves the highest ex-ante Sharpe:*

$$\gamma^* := \arg\max_{\gamma \in \Gamma} \text{SR}(\boldsymbol{w}^*(\gamma)), \qquad \boldsymbol{w}_{t+1}^* := \boldsymbol{w}^*(\gamma^*). \tag{27}$$

In practice we use 50 log-spaced values $\gamma \in [10^{-2}, 10^3]$.

7.3 Deriving the KKT conditions

The Lagrangian of (23) (ignoring the ℓ_1 and box constraints for clarity) is

$$\mathcal{L}(\boldsymbol{w}, \lambda) = (\boldsymbol{\mu}_{t+1}^B - r_f \mathbf{1})^\top \boldsymbol{w} - \gamma \boldsymbol{w}^\top \Sigma_{\text{final}} \boldsymbol{w} - \lambda(\mathbf{1}^\top \boldsymbol{w} - 1). \tag{28}$$

Setting $\nabla_{\boldsymbol{w}} \mathcal{L} = 0$:

$$\boldsymbol{\mu}_{t+1}^B - r_f \mathbf{1} - 2\gamma \Sigma_{\text{final}} \boldsymbol{w} - \lambda \mathbf{1} = 0 \quad \Longrightarrow \quad \boldsymbol{w} = \frac{1}{2\gamma} \Sigma_{\text{final}}^{-1} (\boldsymbol{\mu}_{t+1}^B - r_f \mathbf{1} - \lambda \mathbf{1}). \tag{29}$$

Substituting into $\mathbf{1}^\top \boldsymbol{w} = 1$ and solving for λ:

$$\lambda = \frac{\mathbf{1}^\top \Sigma_{\text{final}}^{-1} (\boldsymbol{\mu}_{t+1}^B - r_f \mathbf{1}) - 2\gamma}{\mathbf{1}^\top \Sigma_{\text{final}}^{-1} \mathbf{1}}. \tag{30}$$

This is the classical two-fund separation result. The additional ℓ_1 and box constraints are handled by the QP solver (they add inequality constraints and corresponding dual variables, but the problem remains convex).

8 Volatility Targeting

After allocation, we scale returns to a target annualised volatility $\sigma^* = 0.15$ (15%):

$$r_t^{\text{VT}} := r_t^{\text{raw}} \cdot \ell_t, \qquad \ell_t := \min\left(\frac{\sigma^*}{\hat{\sigma}_t^{\text{port}}},\ \ell_{\max} \right), \tag{31}$$

where $\hat{\sigma}_t^{\text{port}} = \text{std}(\{r_s^{\text{raw}}\}_{s=t-25}^t) \cdot \sqrt{52}$ and $\ell_{\max} = 2.5$.

Remark 8.1. *This ensures the backtest is comparable across regimes. During calm periods, leverage is increased (up to* $2.5\times$*); during crises, exposure is cut. The 26-week lookback means the adjustment is smooth and avoids whipsawing.*

9 Optimal Transport for Order Execution

We now have a target portfolio $\boldsymbol{w}_{t+1}^*$ and a current portfolio $\boldsymbol{w}_t$. The rebalancing trade $\Delta \boldsymbol{w} = \boldsymbol{w}_{t+1}^* - \boldsymbol{w}_t$ incurs transaction costs. The question is: *what is the cheapest way to rearrange the current portfolio into the target?*

This is precisely a **mass transportation problem**.

9.1 The Kantorovich formulation

Think of the current weights $\boldsymbol{w}_t$ as a "pile of sand" distributed across n assets, and the target $\boldsymbol{w}_{t+1}^*$ as the desired landscape. A *transport plan* $T \in \mathbb{R}_{\geq 0}^{n \times n}$ specifies how much weight to move from asset i to asset j: T_{ij} is the mass shipped from i to j.

Definition 9.1 (Transport polytope)**.** *The set of valid plans:*

$$\Pi(\boldsymbol{a}, \boldsymbol{b}) := \{T \geq 0 : T\mathbf{1} = \boldsymbol{a},\ T^\top \mathbf{1} = \boldsymbol{b}\}, \tag{32}$$

where $\boldsymbol{a} := |\boldsymbol{w}_t| / \||\boldsymbol{w}_t|\|_1$ *and* $\boldsymbol{b} := |\boldsymbol{w}_{t+1}^*| / \||\boldsymbol{w}_{t+1}^*|\|_1$ *are the normalised absolute weights (probability distributions on assets).*

Definition 9.2 (Cost matrix)**.** *The cost of shipping one unit from i to j:*

$$C_{ij} := \begin{cases} 0 & i = j, \\ |c_i| + |c_j| & i \neq j, \end{cases} \tag{33}$$

where c_i is the proportional transaction cost for asset i (default: 10 bps). The $i = j$ case is free because holding an asset costs nothing; the $i \neq j$ case is the cost of selling i and buying j.

Definition 9.3 (Kantorovich optimal transport)**.**

$$T^* := \arg\min_{T \in \Pi(\boldsymbol{a},\boldsymbol{b})} \langle C, T \rangle = \arg\min_{T \in \Pi(\boldsymbol{a},\boldsymbol{b})} \sum_{i,j} C_{ij} T_{ij}. \tag{34}$$

This is a linear programme and can be solved exactly. However, for larger n or frequent rebalancing, we prefer the faster *entropically regularised* version.

9.2 Entropic regularisation

Proposition 9.4 (Regularised problem)**.** *Adding an entropic penalty:*

$$T^*_\varepsilon := \arg\min_{T \in \Pi(\boldsymbol{a},\boldsymbol{b})} \langle C, T \rangle + \varepsilon \sum_{i,j} T_{ij}(\log T_{ij} - 1), \tag{35}$$

with $\varepsilon = 0.05 \cdot \mathrm{median}(\{C_{ij} : i \neq j,\, C_{ij} > 0\})$.

Derivation of the solution structure. Write the Lagrangian with dual variables $\boldsymbol{f} \in \mathbb{R}^n$, $\boldsymbol{g} \in \mathbb{R}^n$ for the marginal constraints:

$$\mathcal{L}(T, \boldsymbol{f}, \boldsymbol{g}) = \sum_{ij} C_{ij} T_{ij} + \varepsilon \sum_{ij} T_{ij}(\log T_{ij} - 1) + \sum_i f_i \Big(a_i - \sum_j T_{ij} \Big) + \sum_j g_j \Big(b_j - \sum_i T_{ij} \Big). \tag{36}$$

Setting $\partial \mathcal{L} / \partial T_{ij} = 0$:

$$C_{ij} + \varepsilon \log T_{ij} - f_i - g_j = 0 \quad \Longrightarrow \quad T_{ij} = \exp\Big(\frac{f_i - C_{ij} + g_j}{\varepsilon} \Big). \tag{37}$$

Define $u_i := e^{f_i/\varepsilon}$, $v_j := e^{g_j/\varepsilon}$, $K_{ij} := e^{-C_{ij}/\varepsilon}$ (the *Gibbs kernel*). Then:

$$\boxed{T^*_{ij} = u_i \, K_{ij} \, v_j.} \tag{38}$$

The vectors $\boldsymbol{u}, \boldsymbol{v}$ are determined by the marginal constraints $T\mathbf{1} = \boldsymbol{a}$ and $T^\top \mathbf{1} = \boldsymbol{b}$, which become:

$$u_i \sum_j K_{ij} v_j = a_i, \qquad v_j \sum_i K_{ij} u_i = b_j. \tag{39}$$

These are solved by alternating normalisation — the Sinkhorn algorithm. □

9.3 Sinkhorn algorithm

Proposition 9.5 (Convergence)**.** *Each Sinkhorn iteration is a coordinate ascent step on the dual problem. The dual objective is strictly concave (since $\varepsilon > 0$), so the iterates converge linearly to the unique optimum at rate $O(e^{-c/\varepsilon})$. In practice, convergence occurs in 10–30 iterations.*

Algorithm 1 Sinkhorn iterations

Input: marginals $\boldsymbol{a}, \boldsymbol{b} \in \Delta^{n-1}$, kernel K, tolerance $\delta = 10^{-8}$
$\boldsymbol{v} \leftarrow \mathbf{1}$
for $\ell = 1, \ldots, 200$ **do**
  $\boldsymbol{u} \leftarrow \boldsymbol{a} \oslash (K\boldsymbol{v})$ *// enforce row marginal*
  $\boldsymbol{v} \leftarrow \boldsymbol{b} \oslash (K^\top \boldsymbol{u})$ *// enforce column marginal*
  if $\|K^\top \boldsymbol{u} \odot \boldsymbol{v} - \boldsymbol{b}\|_\infty < \delta$ **then**
    break
  end if
end for
return $T^* = \operatorname{diag}(\boldsymbol{u})\, K \operatorname{diag}(\boldsymbol{v})$

9.4 Transaction cost and net return

The optimal transport cost is

$$\mathcal{C}^* = \sum_{i \neq j} C_{ij}\, T^*_{ij}, \tag{40}$$

and the net portfolio return is

$$r_t^{\text{raw}} = (\boldsymbol{w}^*_{t+1})^\top \boldsymbol{r}_t - \mathcal{C}^*. \tag{41}$$

10 Performance Metrics

Let $\{r_t\}_{t=1}^T$ be the final (vol-targeted) return series with equity curve $\Pi_t = \prod_{s=1}^t (1 + r_s)$.

$$\text{Total return} := \Pi_T - 1, \tag{42}$$

$$\text{Annualised return} := \Pi_T^{52/T} - 1, \tag{43}$$

$$\text{Annualised vol} := \operatorname{std}(\{r_t\}) \cdot \sqrt{52}, \tag{44}$$

$$\text{Sharpe ratio} := \frac{\text{Ann. return}}{\text{Ann. vol}}, \tag{45}$$

$$\text{Max drawdown} := \min_t \frac{\Pi_t - \max_{s \le t} \Pi_s}{\max_{s \le t} \Pi_s}, \tag{46}$$

$$\text{Calmar ratio} := \frac{\text{Ann. return}}{|\text{MDD}|}, \tag{47}$$

$$\text{Sortino ratio} := \frac{\text{Ann. return}}{\operatorname{std}(\{r_t : r_t < 0\}) \cdot \sqrt{52}}. \tag{48}$$

References

1. Garman, M.B. & Klass, M.J. (1980). On the estimation of security price volatilities from historical data. *J. Business*, 53(1), 67–78.

2. Michaud, R.O. (1989). The Markowitz optimization enigma. *Financial Analysts J.*, 45(1), 31–42.

3. Wolpert, D.H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.

4. Higham, N.J. (2002). Computing the nearest correlation matrix. *IMA J. Numer. Anal.*, 22(3), 329–343.

5. Ledoit, O. & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *J. Multivariate Anal.*, 88(2), 365–411.

6. Cuturi, M. (2013). Sinkhorn distances. *Advances in NIPS*, 26.

7. Peyré, G. & Cuturi, M. (2019). Computational Optimal Transport. *Found. Trends ML*, 11(5–6), 355–607.